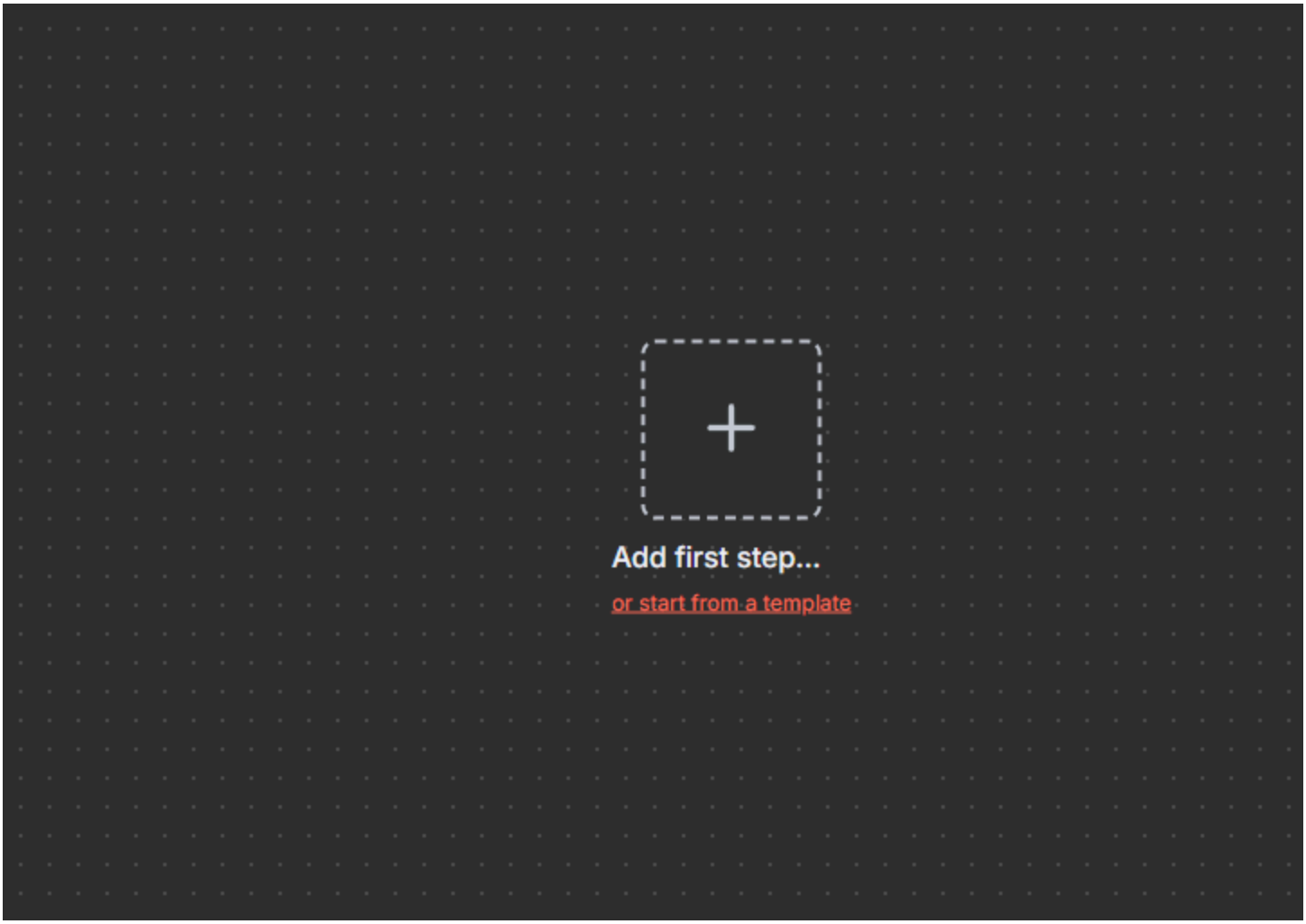
An Introduction to Orchestration Frameworks

What is an agentic orchestration framework
What is N8N, How to set up an account.

Our First N8N Workflow

To understand orchestration, we must first get our hands dirty. We will begin by exploring the core building blocks provided by N8N. When you create a new application in N8N, you are presented with several distinct application types. Think of these as different "species" of AI applications, each evolved for a specific habitat or use case.

We can create a workflow and once done, you will be greeted by a blank **Canvas**.



Understanding the canvas additional docs

https://docs.n8n.io/courses/level-one/chapter-1/

Setting up n8n workflow in google cloud compute

We need to setup a VM to host our instance of N8N. To do this we are going to use google cloud compute. To create a VM instance, follow this guide

Now, onto the docker install, run the following commands:

```
sudo apt update && sudo apt upgrade -y

sudo apt install -y ca-certificates curl gnupg

sudo install -m 0755 -d /etc/apt/keyrings

curl -fsSL https://download.docker.com/linux/debian/gpg | sudo gpg --dearmor -o /etc/apt/

echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] http

sudo apt update

sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-com

sudo systemctl enable --now docker

sudo docker run -it --rm --name n8n -p 5678:5678 docker.n8n.io/n8nio/n8n
```

We want to setup a custom domain name that we can use to reach our n8n instance on our vm. As such, we can use no-ip You can register a free domain here. Once done, we can then configure nginx by creating a confiuration file and using your specific server name, e.g., `carrolln8n.servecounterstrike.com` This allows n8n to be accessed via the website.

First, open a new file using a text editor (here we use nano):
`sudo nano /etc/nginx/sites-available/n8n`

our next step is to run n8n in the background persistently on our machine.

N8N provides great documentation for this
https://docs.n8n.io/hosting/installation/server-setups/docker-compose/#6-create-docker-compose-file

First we need to create an `n8n-compose` directory

```
mkdir n8n-compose
cd n8n-compose
```

Then inside the compose directory we can create an `.env`

```
# DOMAIN_NAME and SUBDOMAIN together determine where n8n will be reachable from
# The top level domain to serve from
DOMAIN_NAME=example.com

# The subdomain to serve from
SUBDOMAIN=n8n

# The above example serve n8n at: https://n8n.example.com

# Optional timezone to set which gets used by Cron and other scheduling nodes
# New York is the default value if not set
GENERIC_TIMEZONE=Europe/Berlin

# The email address to use for the TLS/SSL certificate creation
SSL_EMAIL=user@example.com
```

As an example

```
# Set the first part of your URL here
SUBDOMAIN=carrolln8n

# Set the rest of the domain here
DOMAIN_NAME=servecounterstrike.com
GENERIC_TIMEZONE=Europe/Berlin

# The email address to use for the TLS/SSL certificate creation
SSL_EMAIL=jforden@carrollu.edu
```

we then just need to run the following command

```
docker compose up -d
```

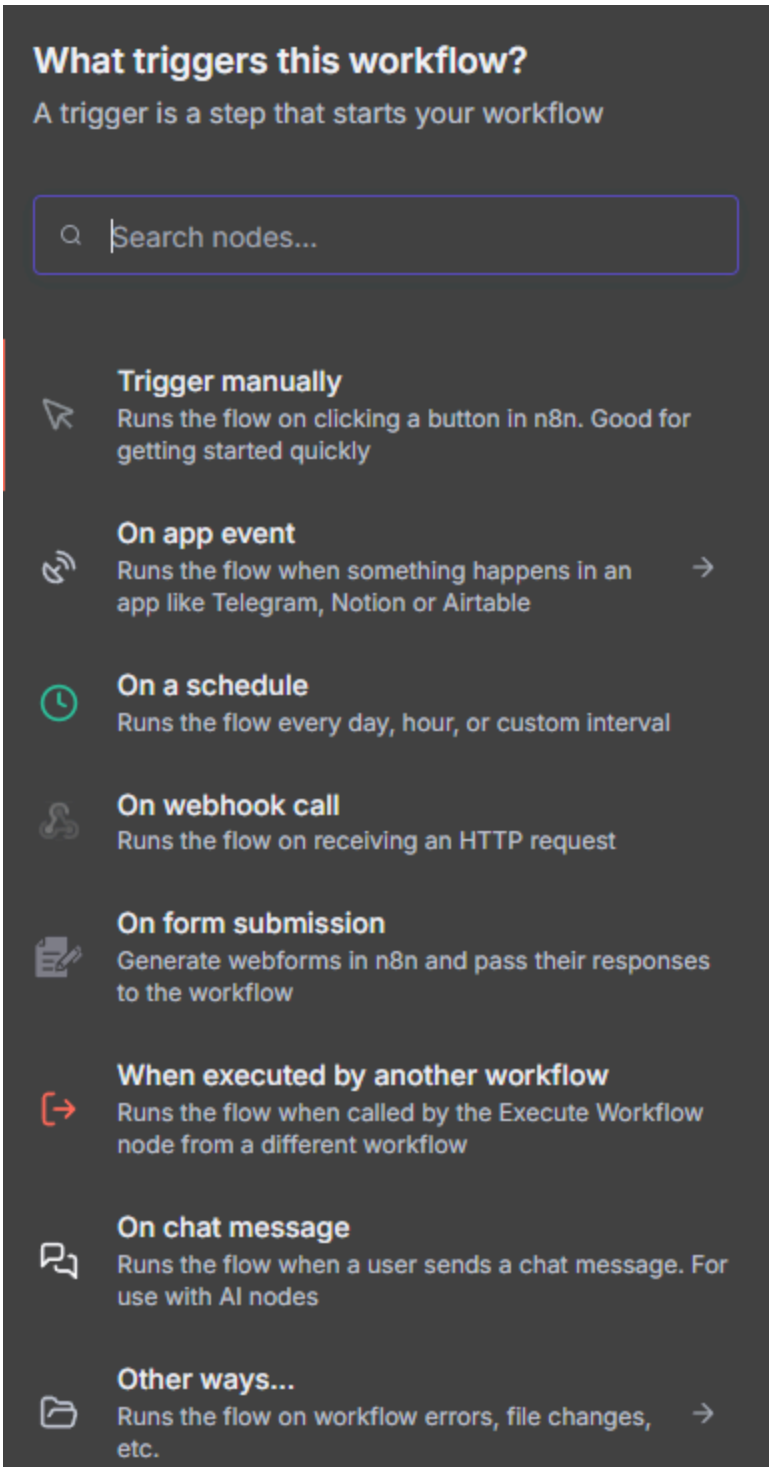
Project 1: The "Hello World" Workflow

For our inaugural project, we will select a **Workflow**. In computer science, a workflow represents a directed graph of tasks—a series of steps that data flows through. This aligns most closely with the traditional definition of "orchestration": coordinating multiple services to achieve a result.

Step 1: Initialization
When you create a blank Workflow, N8N asks you to choose a starting node. This is the *kickoff* point for our entire process.

1. **User Input:**
   - **Concept:** This is a manual trigger. The process remains dormant until a human intervenes—pushing a button, submitting text, or directly moN8Ning the environment.
   - **Analogy:** Think of this like a vending machine. The machine sits idle until you insert a coin and make a selection.
2. **Trigger**
   - **Concept:** This is an automated trigger. It executes based on a schedule (time) or an external event, such as a form submission, a new RSS feed item, or a detected change in a monitored system.
   - **Analogy:** Think of this like an alarm clock or a scheduled news briefing. We can build workflows that automatically wake up when a news report is posted, digest the content, and perform specific actions without any human involvement.

For this first draft, we will select **Trigger Manually**. This allows us to get started quickly and let's us trigger the workflow via a button press. This can be found by selecting the plus icon located on the right side of the screen. The popup you should see is pictured below.

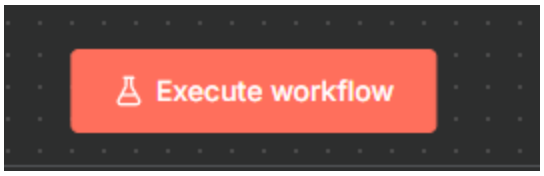


The Canvas is our *workstation*. It allows us to visualize and arrange our individual **nodes**. In graph theory, a node represents a specific point in a network where data is processed. You can think of a node as a station on an assembly line. At each station, a worker (or machine) performs a specific task—pausing to inspect, monitor the product, or passing it quickly to the next stage.

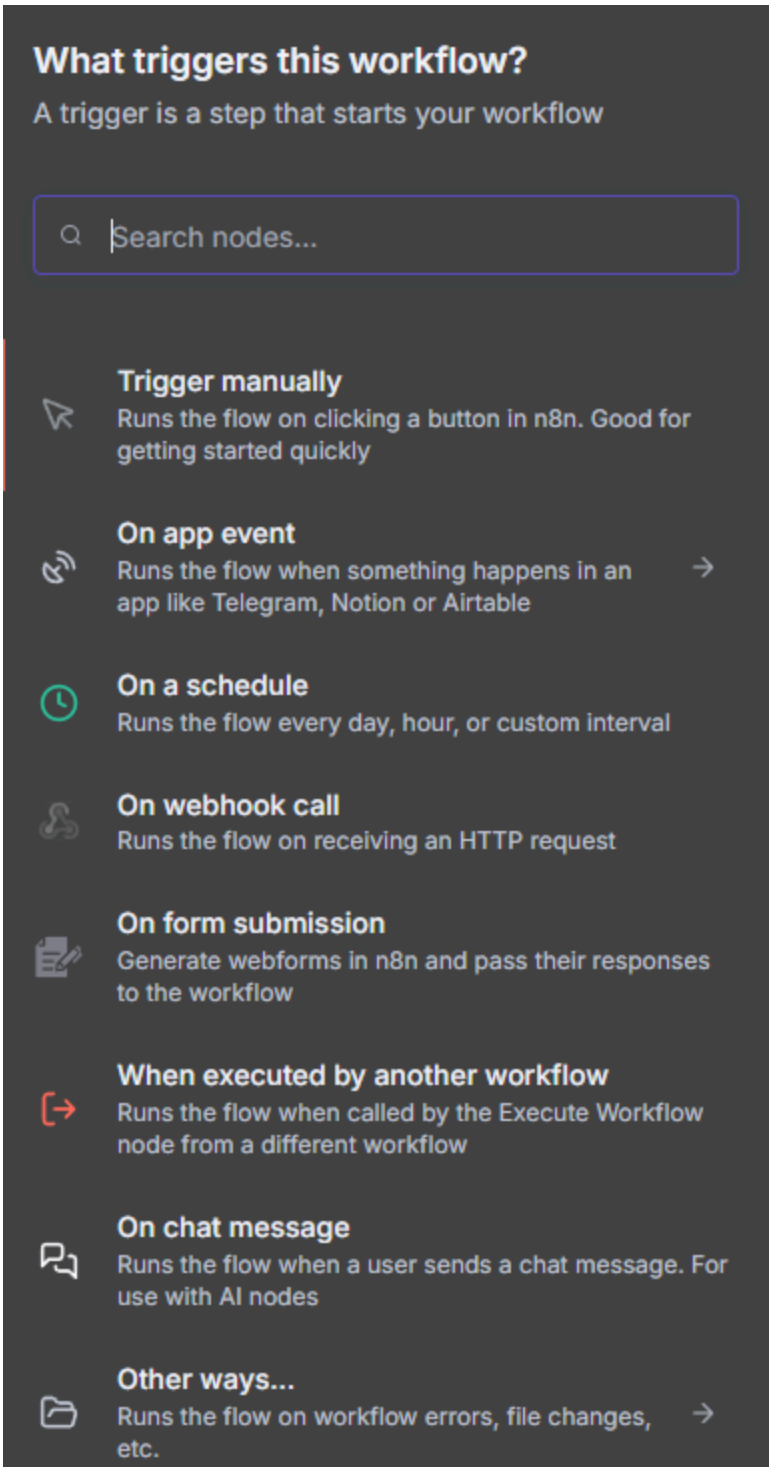
In N8N, nodes are the functional units that give our system power. They allow us to invoke Large Language Models (LLMs), chain information together, access external databases, and execute code. It is through the strategic arrangement of these nodes that we transform our ideas into agentic systems.

- App or Action Nodes add, remove, and edit data; request and send external data; and trigger events in other systems. Refer to the Action nodes library for a full list of these nodes.
- Trigger Nodes start a workflow and supply the initial data. Refer to the Trigger nodes library for a list of trigger nodes.
- Core Nodes can be trigger or app nodes. Whereas most nodes connect to a specific external service, core nodes provide functionality such as logic, scheduling, or generic API calls. Refer to the Core Nodes library for a full list of core nodes.
- Cluster Nodes are node groups that work together to provide functionality in a workflow, primarily for AI workflows. Refer to Cluster nodes for more information.

At the moment, we have a single *start* "USER INPUT" node. We cen *trigger* our run by selecting the "test run" button located towards the top right(pictured below), and you should be greated by a "success" popup denoting that our test run executed without any errors.



Now, clicking *execute workflow* results in a resounding success, yet our workflow doesn't quite do anything grand yet. You can add a new node by selecting the "+" icon that branches directly off from the manual trigger node. For our first example, let's select "core", then "HTTP request"

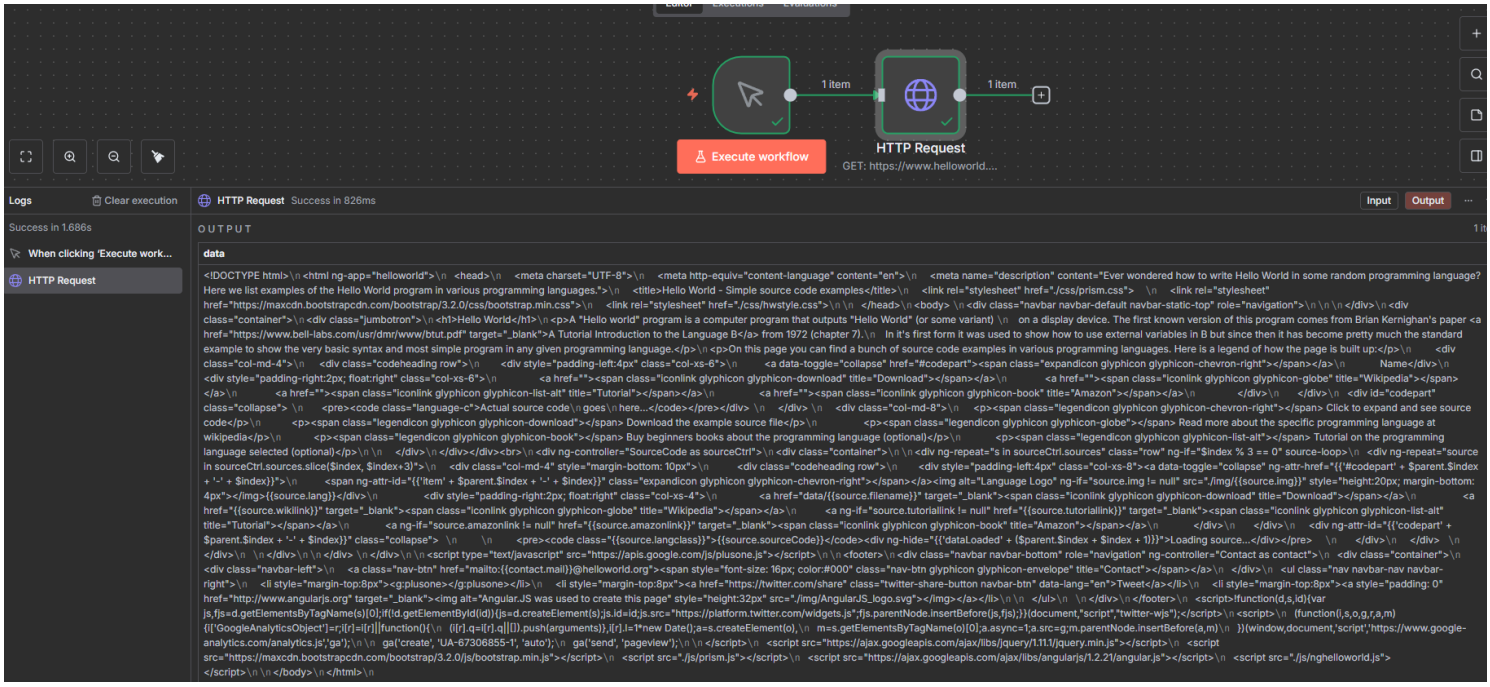


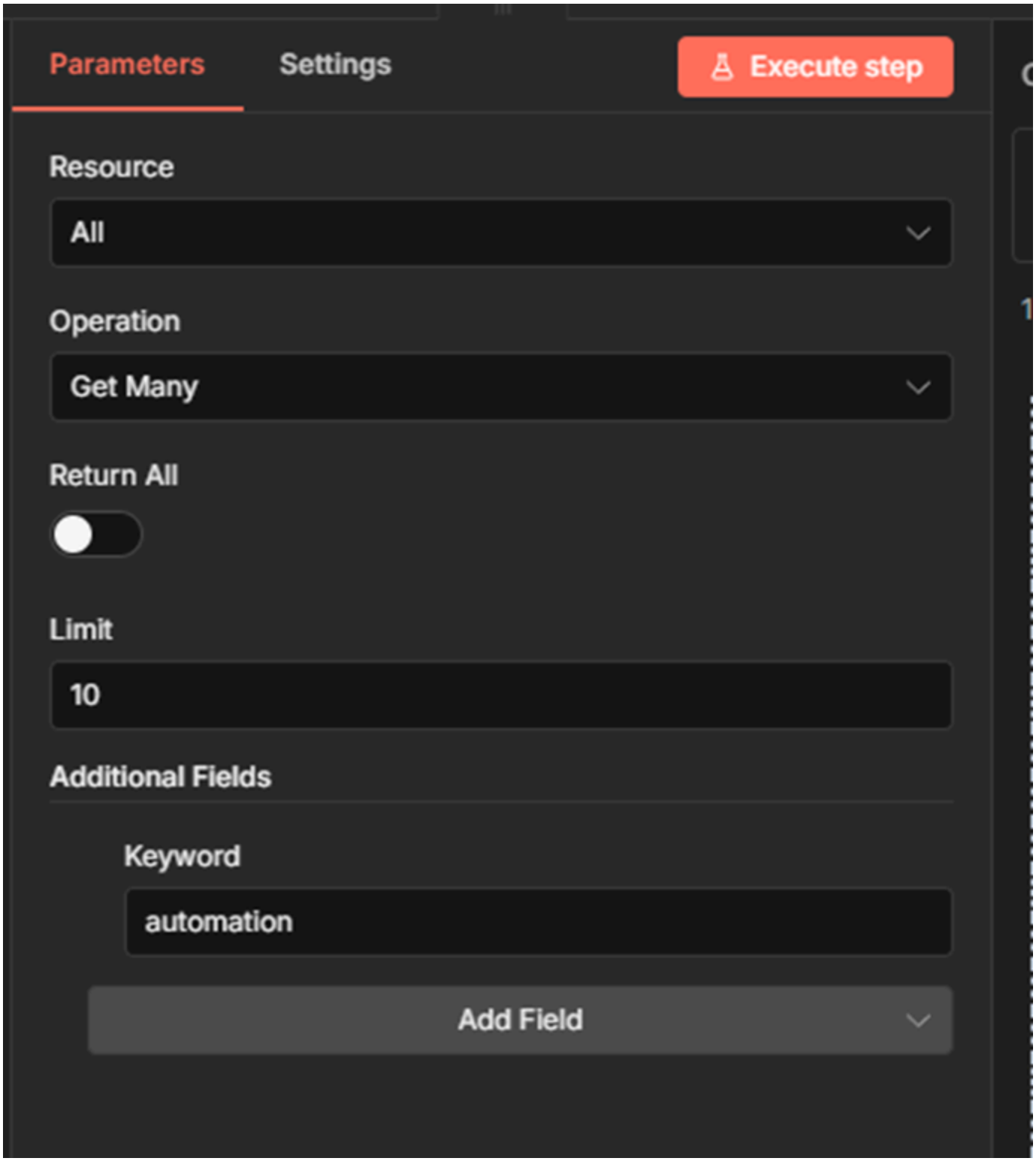
For those unfamiliar with HTTP requests, they are the fundamental protocol of the web. When you type a URL into your browser, you are essentially sending a request to a server asking for information. The most common type is a **GET** request, which, as the name implies, simply asks a server to retrieve specific data and send it back. In our N8N workflow, this node acts as our browser, allowing us to reach out to external services and fetch data to use in our automation.

Let's add the URL `https://www.helloworld.org/` to our node. Once done, closing the popup brings us back to the canvas where we can trigger the workflow again. The screenshot below shows the expected output from the GET request, which may not look incredibly readable to humans. What we are

seeing is HyperText Transfer Markup Language (HTML). HTML is the the raw code that web browsers typically render into the pretty graphics we see on websites. N8N has no such need for this, as typically we will expect agents to parse the important information from this link. If you want, try to change the URL to any website you are familiar with, such as `https://www.carrollu.edu/` and try to see if anything looks familiar in the output.

As of now, we have a very simple N8N workflow going, let's expand this to see a bit more about what N8N can do for us.





Mini Workflow

https://docs.n8n.io/courses/level-one/chapter-2/